

Module A: homework

The Gu pipeline with Fama–French characteristics

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MSCA Doctoral Network Digital Finance · MSCA_DF_10 Introduction to AI for Financial Applications



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Homework: same Gu design, named fundamentals

Gu, S., Kelly, B., & Xiu, D. (2020). *Empirical Asset Pricing via Machine Learning*. *RFS*, 33(5), 2223–2273.

Characteristic names from Fama & French (1993, 2015) *JFE*. This is not a new paper.

Notebook: `notebooks/01b_gu_fama_french.ipynb` (homework, not run on the day)

Sibling: `notebooks/01a_gu_kelly_xiu_simplified.ipynb` (anonymous x_j)

Characteristic OLS (not CAPM)

The classroom **FF** means Fama–French **characteristics**, in the cross-section, ranked:

Name	Spirit
size	SMB
bm	HML
op	RMW
inv	CMA
mom	Carhart (not in FF5)

This is not $r_{it} = \alpha_i + \beta_i r_{m,t}$.

Frozen protocol

The same as notebook 01a :

1. Stock–month panel
2. Calendar train / val / test split
3. Predictive OOS R^2 against a forecast of 0
4. Long–short on $\hat{r}$

Only the **feature story** changes.

Ladder in 01b

Rung	Features
FF3_OLS	size, bm
OLS3_GKX	size, bm, mom (GKX's own OLS-3: $R^2_{\text{OOS}} = 0.16\%$)
FF5_OLS	size, bm, op, inv
OLS-all / Ridge / RF / MLP	named FF + zoo z_0 – z_6
ORACLE_f	true f (DGP cheat)

The 1990s model is the first row, not a competing paper.

Homework questions

1. OLS3_GKX vs FF3_OLS: adding `mom` changes almost nothing. Why? (In this DGP momentum enters only through `size × mom`.)
2. FF3 vs FF5 vs OLS-all: does the zoo help, or does it overfit?
3. Here RF and the MLP can beat the FF baselines because the DGP plants `op2` and `size × mom`. Is that a market fact?

Takeaway

Gu–Kelly–Xiu means: keep the linear factor/characteristic baseline, then add the zoo and ML **under the same OOS rules**.

Back: notebook 01a if you want the anonymous x_j first.